

JAIME UY DE BARON

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[LinkedIn](#)
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About

I am a 4th-year Mechatronics Engineering student and these are my two guiding principles:

- Effect meaningful change.
- Be part of bigger things.

I seek opportunities in a fast-moving environment where I can achieve both of these goals. My passion lies in the biomedical industry as I believe it is an amazing opportunity to do something I love while benefitting the people in my community.

Experience

Design Engineer Intern

Zimmer Biomet, Christchurch, Nov 25 - Feb 26

- Developed standardized reference tools to enhance custom hip implant design, focusing on flange placement and screw trajectory for minimal-incision approaches.
- Designed and implemented modelling methods in closed-source software, integrating tools seamlessly into existing workflows to provide validated guidance on screw trajectories and flange positioning.
- Analyzed 400 total hip arthroplasty cases to quantify average skin depth across posterior, iliofemoral, and anterolateral approaches, informing surgical accessibility constraints.
- Conducted 3D scanning and mesh analysis of post-print titanium implants to assess underbuild; led a study of 40+ cases to define an evidence-based underbuild tolerance with manufacturing partners.
- Gained exposure to orthopedic implant design principles (hip, knee, shoulder) and knee alignment methodologies.

PhD Support Researcher

University of Otago, Christchurch, Nov 24 - Feb 25

- Developed a methodology to improve tumour margin assessment by correlating high-resolution histology with low-resolution CT imaging.
- Designed minimally invasive measurement techniques using rapid Python-based calculations integrated with existing tools (band saw, vernier calipers).
- Proposed and validated a novel method demonstrating superior accuracy (mean error 0.02 mm) compared to prior techniques (0.12 mm and 0.06 mm), tested on canine samples.
- Authored a scientific manuscript detailing methodology and results; currently in revision phases with pending acceptance into the Australian College of Physicists, Engineers, and Scientists in Medicine journal.

Assistant Analyst

Forsyth Barr Investment Bank, Dec 23 - Present

- Author of the Morning News Summary, a published briefing on overnight global equity activity, analysing earnings releases, macro drivers, and stock price movements.
- Conduct research across US, European, Asian, and Australian markets; synthesise analyst reports on key holdings and deliver concise briefings for an 8:00 am adviser meeting.
- Maintain and optimise complex Excel-based databases tracking domestic equities and historical share price data for adviser use and dissemination to external database managers.

Education

Bachelor of Mechatronics Engineering

University of Canterbury, 2023 - 2026

GPA 8.20/9.00

UC Hiranga Scholarship

High School

Auckland Grammar School, 2018 - 2022

Cambridge A Levels (91% Physics, 94% Business, 88% Economics)

Prizes in Business Studies, Economics, Spanish

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Projects

Automated Reprocessing of Medical Devices

Team Lead | Automation Engineer

- Leading a team of five to design and build a fully autonomous medical device reprocessing system that eliminates numerous hours of manual labour and cost.
- Pioneering an automated singulator for disinfectant wipes while maintaining its moisture content and structural integrity for accurate simulation of clinical conditions.

RoboCup Autonomous Robotics Competition

Team Lead | Systems Engineering | Embedded Software

- Led a team of three to design and build a fully autonomous robot capable of obstacle avoidance, target detection, collection, and return-to-base navigation.
- Designed the main control software across 10+ modularized files to improve maintainability and rapid debugging.
- Designed and manufactured custom mechanical components, including laser-cut acrylic plates and a redesigned sensor mounting system.
- Integrated mechanical subsystems with sensor identification and timing-critical control logic.
- Iteratively debugged and refined navigation logic through extensive situational testing.
- Delivered a competition-ready system with one of the highest autonomous return and target collection success rates.

Technical Skills

- High attention to detail, especially in meticulous documentation of thoughts and project progress.
- Program logic (Python, Ladder Logic and Structured Text, C++).
- Turning manufacturing (2nd year project) and sheet metal manufacturing (3rd year project).
- Data-processing expertise (Python and Excel).
- Laser cutting and 3D printing.
- Embedded system design and implementation.
- Parametric CAD packages: Sketchup, SolidWORKS and organic CAD packages: Geomagic Freeform Plus.
- PCB Design: Altium Designer, KiCAD.
- CT Scan Segmentation: 3DSlicer, Mimics.

References

Available on request.

Leadership

President (Former Treasurer), Mechatronics Society
(2024 - Present)

- Provide strategic and financial leadership; expanded event portfolio under <\$500 base budget.
- Founded and led the nationwide UC Engineering Challenge, securing \$6,000 in sponsorship; 320 participants, 70 technical submissions.
- Launched initiatives including Internships 101, Meet Mechatronics, and the UC Pitching Competition.

Treasurer, Lads Without Labels (2024 - 2025)

- Managed \$200,000+ in assets; oversaw GST, charity filings, budgeting, and compliance.
- Contributed to brand growth and strategic expansion to additional universities.

Marketing Manager (Former Treasurer), Student Volunteer Army (2024 - Present)

- Managed \$30,000 in assets and finance operations prior to transitioning into marketing leadership.
- Coordinated disaster relief operations across Nelson, Hawke's Bay, and Banks Peninsula.
- Directed marketing for large-scale initiatives including the Big Give (500 volunteers mobilised).

Proudest Achievements

- Raised \$2,400 for flood victims in Wairoa.
- Ran 72 km across 3 days for men's mental health.
- Australian journal submission in my 2nd year of university.
- Competed in and won the DOC Takahē Challenge in 2024.